HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed and executed **system-level test specifications** for automotive camera and mechatronic ECU systems at HCL, ensuring **functional and performance requirement coverage** aligned with automotive standards; developed **test methods** using **Python** and **C++** scripting to validate sensor integration and feature behavior across bench and vehicle environments, achieving **98% test traceability** to system requirements.
- Planned and implemented **test automation capability** following platform standards, building **automated test harnesses** in **NI TestStand** and **NI LabVIEW** for stress and robustness testing of automotive software features; reduced manual test execution time by **45%** while maintaining comprehensive coverage of system functional specifications.
- Reviewed **system requirements and architecture design** to assess testability and plan **test coverage** in collaboration with System Engineers; specified **test environment requirements** for Bench and HIL environments, contributing to test tool developer specifications and ensuring on-time delivery of **system validation targets** across the project lifecycle.
- Executed **trace analysis and troubleshooting** on automotive ECU systems and ADAS software features to determine root cause of validation failures; documented defect findings with technical depth, communicated issues to customer teams and engineering stakeholders, and validated fixes, reducing system-level issue resolution time by **35%**.
- Maintained comprehensive **test results, test logs, and test reports** throughout project lifecycle using **Python** data processing and report automation; monitored progression of planned testing and demonstrated test completion status with **100% traceability** of test cases to functional and performance requirements, supporting UAT and post-go-live validation phases.
- Supported cross-functional collaboration with customer teams on joint testing events and technical workshops; contributed to presentations and reports detailing system validation status, detected defects, and overall execution against automotive industry standards; developed and maintained professional relationships within test teams and across Systems Engineering and Function Owner departments.

SKILLS

Programming & Scripting – Python, C++, C, C#, NI LabVIEW, NI TestStand, shell scripting.

Test Engineering & Validation – System-level testing, bench testing, HIL (Hardware-in-Loop) testing, vehicle testing, test specification design, test automation, stress testing, robustness testing, test coverage planning, trace analysis, root cause analysis.

Automotive Systems & Standards – ADAS (Advanced Driver Assistance Systems), mechatronic ECU systems, automotive camera systems, automotive software feature validation, automotive industry standards compliance.

Systems Development & Requirements – Systems Development Life Cycle (SDLC), requirements traceability, test traceability matrix, architecture design review, testability assessment, functional requirements validation, performance requirements validation.

Test Tools & Platforms – NI LabVIEW, NI TestStand, Jenkins CI/CD pipelines, Git version control, Jira defect tracking.

Data & Reporting – SQL (PostgreSQL, MySQL, Oracle), test log analysis, test report generation, data processing and automation, technical documentation.

Soft Skills & Collaboration – Technical communication with customer teams, cross-functional collaboration, UAT support, defect reporting, presentation and report writing, Agile/Scrum methodologies.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Automotive ADAS Camera System Validation & Test Automation Platform (HCL Technologies)

- Designed and implemented a comprehensive **system-level test specification** for a multi-sensor ADAS camera system supporting autonomous vehicle features; created **test methods** in **Python** and **NI LabVIEW** to validate functional requirements (lane detection, object recognition, pedestrian alert) and performance metrics (latency <50ms, accuracy >95%) across bench and vehicle environments.
- Developed automated test harnesses in **NI TestStand** to execute **stress and robustness testing** on camera ECU firmware and automotive software features; integrated sensor simulation models and vehicle data playback, reducing manual test cycles by **45%** and enabling continuous validation across project milestones.
- Established **test traceability matrix** linking all **980+ test cases** to system functional and performance requirements; maintained **test results, logs, and reports** using Python-based automation, achieving **100% requirement coverage** and enabling real-time test status reporting to customer teams and cross-functional engineering stakeholders.
- Planned and executed **HIL (Hardware-in-Loop) and vehicle testing campaigns** to validate ADAS feature behavior in realistic driving scenarios; conducted **trace analysis** on vehicle CAN bus logs to identify root causes of validation failures, reducing system-level issue resolution time by **38%** and supporting UAT sign-off within project timeline.
- Collaborated with System Engineers and Function Owners to assess **architecture testability** and defined **test environment requirements** for bench, HIL, and vehicle configurations; contributed to technical workshops with automotive customer teams, communicating validation status, defect findings, and recommended design improvements to support production readiness.

Enterprise Application Test Automation & Production Monitoring (HCL Technologies)

- Designed and executed **system validation test plans** for enterprise Java/Spring Boot applications supporting banking operations; developed **test automation** using **Python** and **CI/CD pipelines** (Jenkins) to validate functional correctness and performance under load, achieving **92% automated test coverage**.
- Implemented **trace analysis** and root cause investigation workflows using **Splunk, Dynatrace, and CloudWatch** to identify and troubleshoot production incidents; documented technical defect reports with detailed logs and execution traces, reducing mean time to resolution (MTTR) by **40%** and maintaining **99.5% SLA compliance** across mission-critical systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)